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MIMO
AI for Communication
Communication for Computing

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Multicell MIMO communications relying on intelligent reflecting surfaces C Pan, H Ren, K Wang, W Xu, M El Kashlan, A Nallanathan, L Hanzo IEEE transactions on wireless communications 19 (8), 5218-5233	1220	2020
Low-complexity hybrid precoding in massive multiuser MIMO systems L Liang, W Xu, X Dong IEEE Wireless Communications Letters 3 (6), 653-656	889	2014
Edge learning for B5G networks with distributed signal processing: Semantic communication, edge computing, and wireless sensing W Xu, Z Yang, DWK Ng, M Levorato, YC Eldar, M Debbah IEEE journal of selected topics in signal processing 17 (1), 9-39	735	2023
Secrecy rate maximization for intelligent reflecting surface assisted multi-antenna communications H Shen, W Xu, S Gong, Z He, C Zhao IEEE Communications Letters 23 (9), 1488-1492	625	2019
Multi-agent deep reinforcement learning-based trajectory planning for multi-UAV assisted mobile edge computing L Wang, K Wang, C Pan, W Xu, N Aslam, L Hanzo IEEE Transactions on Cognitive Communications and Networking 7 (1), 73-84	542	2020
Resource allocation for D2D-enabled vehicular communications L Liang, GY Li, W Xu IEEE Transactions on Communications 65 (7), 3186-3197	490	2017
Energy-efficient wireless communications with distributed reconfigurable intelligent surfaces Z Yang, M Chen, W Saad, W Xu, M Shikh-Bahaei, HV Poor, S Cui IEEE Transactions on Wireless Communications 21 (1), 665-679	394	2021
Full-duplex communication for ISAC: Joint beamforming and power optimization Z He, W Xu, H Shen, DWK Ng, YC Eldar, X You IEEE Journal on Selected Areas in Communications 41 (9), 2920-2936	380	2023
Deep reinforcement learning based dynamic trajectory control for UAV-assisted mobile edge computing L Wang, K Wang, C Pan, W Xu, N Aslam, A Nallanathan IEEE Transactions on Mobile Computing 21 (10), 3536-3550	364	2021
5G cellular user equipment: From theory to practical hardware design Y Huo, X Dong, W Xu	299	2017

TITLE	CITED BY	YEAR
IEEE access 5, 13992-14010		
Energy efficient UAV communication with energy harvesting Z Yang, W Xu, M Shikh-Bahaei IEEE Transactions on Vehicular Technology 69 (2), 1913-1927	298	2019
On the optimality of power allocation for NOMA downlinks with individual QoS constraints Z Yang, W Xu, C Pan, Y Pan, M Chen IEEE Communications Letters 21 (7), 1649-1652	255	2017
Deep-learning-based joint resource scheduling algorithms for hybrid MEC networks F Jiang, K Wang, L Dong, C Pan, W Xu, K Yang IEEE Internet of Things Journal 7 (7), 6252-6265	245	2019
Spectral and energy efficiency of IRS-assisted MISO communication with hardware impairments S Zhou, W Xu, K Wang, M Di Renzo, MS Alouini IEEE wireless communications letters 9 (9), 1366-1369	239	2020
Sum-rate maximization of uplink rate splitting multiple access (RSMA) communication Z Yang, M Chen, W Saad, W Xu, M Shikh-Bahaei IEEE Transactions on Mobile Computing 21 (7), 2596-2609	230	2020
Standard analysis for transfer delay in CTCS - 3 Y Cao, L Ma, S Xiao, X Zhang, W Xu Chinese Journal of Electronics 26 (5), 1057-1063	224	2017
Energy efficient resource allocation in machine-to-machine communications with multiple access and energy harvesting for IoT Z Yang, W Xu, Y Pan, C Pan, M Chen IEEE Internet of Things Journal 5 (1), 229-245	222	2017
MIMO channel information feedback using deep recurrent network C Lu, W Xu, H Shen, J Zhu, K Wang IEEE Communications Letters 23 (1), 188-191	178	2018
Joint altitude, beamwidth, location, and bandwidth optimization for UAV-enabled communications Z Yang, C Pan, M Shikh-Bahaei, W Xu, M Chen, M Elkashlan, ... IEEE Communications Letters 22 (8), 1716-1719	178	2018
Intelligent reflection enabling technologies for integrated and green Internet-of-Everything beyond 5G: Communication, sensing, and security W Shi, W Xu, X You, C Zhao, K Wei IEEE Wireless Communications 30 (2), 147-154	177	2022



附件一. 文献类型及归属情况统计表

SCIE 数据库

文献类型	文献数量	第一作者	通讯作者（第一作者除外）
Article	72	6	58
Editorial Material	3	3	0
Review	1	0	1

附件二. 文献收录引用情况表

一. SCIE 收录论文在 WOS 核心合集的引用次数表

序号	题名	论文出处	发表时间	被引频次总计 (WOS)	实际被引频次	他引频次
1.	Energy-Efficient Edge Inference in Integrated Sensing, Communication, and Computation Networks	IEEE JOURNAL ON SELECTED AREAS IN COMMUNICATIONS	OCT 2025	2	2	1
2.	Edge Learning for B5G Networks With Distributed Signal Processing: Semantic Communication, Edge Computing, and Wireless Sensing	IEEE JOURNAL OF SELECTED TOPICS IN SIGNAL PROCESSING	JAN 2023	432	432	279
3.	Disentangled Representation Learning Empowered CSI Feedback Using Implicit Channel Reciprocity in FDD Massive MIMO	IEEE TRANSACTIONS ON WIRELESS COMMUNICATIONS	OCT 2024	8	8	4
4.	Energy-Efficient Hybrid Beamfocusing for Near-Field Integrated Sensing and Communication	IEEE TRANSACTIONS ON SIGNAL PROCESSING	2025	0	0	0
5.	Cooperative Reflection and Synchronization Design for Distributed Multiple-RIS Communications	IEEE JOURNAL OF SELECTED TOPICS IN SIGNAL PROCESSING	AUG 2022	35	35	28
6.	Toward ubiquitous and intelligent 6G networks: from architecture to technology	SCIENCE CHINA-INFORMATION SCIENCES	MAR 2023	47	47	16
7.	Spectral and Energy Efficiency of Multi-Pair Massive MIMO Relay Network With Hybrid Processing	IEEE TRANSACTIONS ON COMMUNICATIONS	SEP 2017	62	61	44
8.	Robust Beamforming With Partial Channel State Information for Energy Efficient Networks	IEEE JOURNAL ON SELECTED AREAS IN COMMUNICATIONS	DEC 2015	65	65	46
9.	Optimized One-Way Relaying Strategy With Outdated CSI Quantization for Spatial Multiplexing	IEEE TRANSACTIONS ON SIGNAL PROCESSING	AUG 2012	23	23	18
10.	MIMO Relaying Broadcast Channels With Linear Precoding and Quantized Channel State Information Feedback	IEEE TRANSACTIONS ON SIGNAL PROCESSING	OCT 2010	59	58	39



11.	Over-the-Air Multitask Federated Learning via Cost-Effective Hybrid Beamforming	IEEE TRANSACTIONS ON SIGNAL PROCESSING	2025	3	3	2
12.	Wireless Federated Learning Over Resource-Constrained Networks: Digital Versus Analog Transmissions	IEEE TRANSACTIONS ON WIRELESS COMMUNICATIONS	OCT 2024	21	21	15
13.	Full-Duplex Communication for ISAC: Joint Beamforming and Power Optimization	IEEE JOURNAL ON SELECTED AREAS IN COMMUNICATIONS	SEP 2023	195	195	178
14.	MSE-Based Training and Transmission Optimization for MIMO ISAC Systems	IEEE TRANSACTIONS ON SIGNAL PROCESSING	2024	17	17	16
15.	Low-Complexity Hybrid Precoding in Massive Multiuser MIMO Systems	IEEE WIRELESS COMMUNICATIONS LETTERS	DEC 2014	584	574	530
16.	Energy-Efficient Wireless Communications With Distributed Reconfigurable Intelligent Surfaces	IEEE TRANSACTIONS ON WIRELESS COMMUNICATIONS	JAN 2022	243	243	225
17.	A Framework of Arithmetic-Level Variable Precision Computing for In-Memory Architecture: Case Study in MIMO Signal Processing	IEEE TRANSACTIONS ON MOBILE COMPUTING	JAN 2026	0	0	0
18.	Distributed Multiuser Precoding via Unfolding WMMSE for Multicell MIMO Downlinks	IEEE TRANSACTIONS ON COGNITIVE COMMUNICATIONS AND NETWORKING	AUG 2025	1	1	1
19.	On privacy, security, and trustworthiness in distributed wireless large AI models	SCIENCE CHINA-INFORMATION SCIENCES	JUL 2025	3	3	3
20.	Byzantine-Resilient Over-the-Air Federated Learning Under Zero-Trust Architecture	IEEE JOURNAL ON SELECTED AREAS IN COMMUNICATIONS	JUN 2025	2	2	0
21.	Amplitude-Phase-Time Block Modulation for Resisting Nonlinear Amplification and Its Application for Energy-Efficient Wireless Communications	IEEE TRANSACTIONS ON COMMUNICATIONS	APR 2025	1	1	0
22.	Hybrid Beamforming for Millimeter-Wave Massive Grant-Free Transmission	IEEE TRANSACTIONS ON COMMUNICATIONS	MAR 2025	7	7	2
23.	Combating Interference for Over-the-Air Federated Learning: A Statistical Approach via RIS	IEEE TRANSACTIONS ON SIGNAL PROCESSING	2025	10	10	4
24.	Asymmetric PoolCsiNet With Parameter-Free Encoder at UE for CSI Feedback	IEEE TRANSACTIONS ON COMMUNICATIONS	DEC 2024	0	0	0



51.	Beamforming Optimization for IRS-Aided Communications With Transceiver Hardware Impairments	IEEE TRANSACTIONS ON COMMUNICATIONS	FEB 2021	96	96	84
52.	Beamforming Design for Multiuser Transmission Through Reconfigurable Intelligent Surface	IEEE TRANSACTIONS ON COMMUNICATIONS	JAN 2021	86	86	76
53.	Spectral and Energy Efficiency of IRS-Assisted MISO Communication With Hardware Impairments	IEEE WIRELESS COMMUNICATIONS LETTERS	SEPT 2020	158	157	132
54.	Secure Communication for Spatially Sparse Millimeter-Wave Massive MIMO Channels via Hybrid Precoding	IEEE TRANSACTIONS ON COMMUNICATIONS	FEB 2020	51	51	35
55.	Multicell MIMO Communications Relying on Intelligent Reflecting Surfaces	IEEE TRANSACTIONS ON WIRELESS COMMUNICATIONS	AUG 2020	867	865	726
56.	Monthly Service Prediction for 4G/5G Systems: A Short Time Series-Based Neural Network Solution	IEEE TRANSACTIONS ON COGNITIVE COMMUNICATIONS AND NETWORKING	2026	1	1	1
57.	Robust ISAC Transceiver Beamforming Design Under Low-Resolution AD/DA Converters	IEEE WIRELESS COMMUNICATIONS LETTERS	NOV 2025	0	0	0
58.	Rate Splitting and Beamforming Design for RIS-Assisted RSMA-Enhanced ISAC Networks	IEEE WIRELESS COMMUNICATIONS LETTERS	JUN 2025	5	5	4
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61.	Completion Time Minimization Oriented Cross-Layer Design in Multiuser Communication Networks	IEEE WIRELESS COMMUNICATIONS LETTERS	JAN 2025	0	0	0
62.	Learning Power Allocation and Channel Estimation for Superimposed Pilot-Assisted Multiuser MIMO	IEEE WIRELESS COMMUNICATIONS LETTERS	NOV 2024	2	2	2
63.	Enhancing Wideband Multiuser MIMO Uplink Using Superimposed Pilots: Joint Receiver Design	IEEE WIRELESS COMMUNICATIONS LETTERS	APR 2024	3	3	2
64.	Low-Complexity Channel Estimation for Extremely Large-Scale MIMO in Near Field	IEEE WIRELESS COMMUNICATIONS LETTERS	MAR 2024	28	28	27
65.	GoMORE: Global Model Reuse for Resource-Constrained Wireless Federated Learning	IEEE WIRELESS COMMUNICATIONS LETTERS	SEP 2023	22	22	14



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1 A Joint Learning and Communications Framework for Federated Learning Over Wireless Networks

1,135 Citations37 References

Chen, MZ; Yang, ZH; (...); Cui, SG

Jan 2021 | IEEE TRANSACTIONS ON WIRELESS COMMUNICATIONS 20(1), pp.269-283

Enriched Cited References

In this article, the problem of training federated learning (FL) algorithms over a realistic wireless network is studied. In the considered model, wireless users execute an FL algorithm while training their local FL models using their own data and transmitting the trained local FL models to a base station (BS) that generates a global FL mo ... Show more

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2 Wireless Communications With Reconfigurable Intelligent Surface: Path Loss Modeling and Experimental Measurement

1,024 Citations47 References

Tang, WK; Chen, MZ; (...); Cui, TJ

Jan 2021 | IEEE TRANSACTIONS ON WIRELESS COMMUNICATIONS 20(1), pp.421-439

Reconfigurable intelligent surfaces (RISs) comprised of tunable unit cells have recently drawn significant attention due to their superior capability in manipulating electromagnetic waves. In particular, RIS-assisted wireless communications have the great potential to achieve significant performance improvement and coverage enhancen ... Show more

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3 Multicell MIMO Communications Relying on Intelligent Reflecting Surfaces

894 Citations46 References

Pan, CH; Ren, H; (...); Hanzo, L

Aug 2020 | IEEE TRANSACTIONS ON WIRELESS COMMUNICATIONS 19(8), pp.5218-5233

Intelligent reflecting surfaces (IRSs) constitute a disruptive wireless communication technique capable of creating a controllable propagation environment. In this paper, we propose to invoke an IRS at the cell boundary of multiple cells to assist the downlink transmission to cell-edge users, whilst mitigating the inter-cell interferen ... Show more

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4 Federated Learning via Over-the-Air Computation

870 Citations55 References

Yang, K; Jiang, T; (...); Ding, Z

Mar 2020 | IEEE TRANSACTIONS ON WIRELESS COMMUNICATIONS 19(3), pp.2022-2035

The stringent requirements for low-latency and privacy of the emerging high-stake applications with intelligent devices such as drones and smart vehicles make the cloud computing inapplicable in these scenarios. Instead, edge machine learning becomes increasingly attractive for performing training and inference directly at net ... Show more

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5 Energy Efficient Federated Learning Over Wireless Communication Networks

841 Citations40 References

Yang, ZH; Chen, MZ; (...); Shikh-Bahaei, M

Mar 2021 | IEEE TRANSACTIONS ON WIRELESS COMMUNICATIONS 20(3), pp.1935-1949

In this paper, the problem of energy efficient transmission and computation resource allocation for federated learning (FL) over wireless communication networks is investigated. In the considered model, each user exploits limited local computational resources to train a local FL model with its collected data and, then, sends th ... Show more

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6 Weighted Sum-Rate Maximization for Reconfigurable Intelligent Surface Aided Wireless Networks

836 Citations52 References

Guo, HY; Liang, YC; (...); Larsson, EG

May 2020 | IEEE TRANSACTIONS ON WIRELESS COMMUNICATIONS 19(5), pp.3064-3076

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1 Optimal LAP Altitude for Maximum Coverage



Al-Hourani, A; Kandeepan, S and Lardner, S

Dec 2014 | IEEE WIRELESS COMMUNICATIONS LETTERS 3(6), pp.569-572

Low-altitude aerial platforms (LAPs) have recently gained significant popularity as key enablers for rapid deployable relief networks where coverage is provided by onboard radio heads. These platforms are capable of delivering essential wireless communication for public safety agencies in remote areas or during the aftermath of natural ...

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Citations

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References

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2 Power of Deep Learning for Channel Estimation and Signal Detection in OFDM Systems



Ye, H; Li, GY and Juang, BH

Feb 2018 | IEEE WIRELESS COMMUNICATIONS LETTERS 7(1), pp.114-117

This letter presents our initial results in deep learning for channel estimation and signal detection in orthogonal frequency-division multiplexing (OFDM) systems. In this letter, we exploit deep learning to handle wireless OFDM channels in an end-to-end manner. Different from existing OFDM receivers that first estimate channel state infor ...

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3 Deep Learning for Massive MIMO CSI Feedback



Wen, CK; Shih, WT and Jin, S

Oct 2018 | IEEE WIRELESS COMMUNICATIONS LETTERS 7(5), pp.748-751

In frequency division duplex mode, the downlink channel state information (CSI) should be sent to the base station through feedback links so that the potential gains of a massive multiple-input multiple-output can be exhibited. However, such a transmission is hindered by excessive feedback overhead. In this letter, we use deep l ...

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4 Intelligent Reflecting Surface Versus Decode-and-Forward: How Large Surfaces are Needed to Beat Relaying?



Björnson, E; Özdogan, Ö and Larsson, EG

Feb 2020 | IEEE WIRELESS COMMUNICATIONS LETTERS 9(2), pp.244-248

The rate and energy efficiency of wireless channels can be improved by deploying software-controlled metasurfaces to reflect signals from the source to the destination, especially when the direct path is weak. While previous works mainly optimized the reflections, this letter compares the new technology with classic decode-and-f ...

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5 Secure Wireless Communication via Intelligent Reflecting Surface



Cui, M; Zhang, GC and Zhang, R

Oct 2019 | IEEE WIRELESS COMMUNICATIONS LETTERS 8(5), pp.1410-1414

An intelligent reflecting surface (IRS) can adaptively adjust the phase shifts of its reflecting units to strengthen the desired signal and/or suppress the undesired signal. In this letter, we investigate an IRS-aided secure wireless communication system where a multi-antenna access point (AP) sends confidential messages to a single-ant ...

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6 3-D Placement of an Unmanned Aerial Vehicle Base Station (UAV-BS) for Energy-Efficient Maximal Coverage



Alzenad, M; El-Keyi, A; (...); Yanikomeroglu, H

Aug 2017 | IEEE WIRELESS COMMUNICATIONS LETTERS 6(4), pp.434-437

Unmanned aerial vehicle mounted base stations (UAV-BSs) can provide wireless services in a variety of scenarios. In this letter, we propose an optimal placement algorithm for UAV-BSs that maximizes the number of covered users using the minimum transmit power. We decouple the UAV-BS deployment problem in the vertical and ...

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7	Intelligent Reflecting Surface-Enhanced OFDM: Channel Estimation and Reflection Optimization Zheng, BX and Zhang, R Apr 2020 IEEE WIRELESS COMMUNICATIONS LETTERS 9(4), pp.518-522 In the intelligent reflecting surface (IRS)-enhanced wireless communication system, channel state information (CSI) is of paramount importance for achieving the passive beamforming gain of IRS, which, however, is a practically challenging task due to its massive number of passive elements without transmitting/receiving capabilities. In this letter, we propose a novel IRS-aided OFDM system, where the IRS is designed to reflect the incident signals by adjusting the phase shift of each element. The proposed system can achieve a significant performance gain compared with the conventional IRS system. The results show that the proposed system can achieve a significant performance gain compared with the conventional IRS system. The results show that the proposed system can achieve a significant performance gain compared with the conventional IRS system. 全文链接 View full text Free Submitted Article From Repository View Full Text on ProQuest ...	741 Citations 8 References Related records
8	Cascaded Channel Estimation for Large Intelligent Metasurface Assisted Massive MIMO He, ZQ and Yuan, XJ Feb 2020 IEEE WIRELESS COMMUNICATIONS LETTERS 9(2), pp.210-214 In this letter, we consider the problem of channel estimation for large intelligent metasurface (LIM) assisted massive multiple-input multiple-output (MIMO) systems. The main challenge of this problem is that the LIM integrated with a large number of low-cost metamaterial antennas can only passively reflect the incident signals by adjusting the phase shift of each element. The proposed system can achieve a significant performance gain compared with the conventional LIM system. The results show that the proposed system can achieve a significant performance gain compared with the conventional LIM system. The results show that the proposed system can achieve a significant performance gain compared with the conventional LIM system. 全文链接 Full Text at Publisher ...	622 Citations 15 References Related records
9	Intelligent Reflecting Surfaces: Physics, Propagation, and Pathloss Modeling Özdoğan, Ö; Björnson, E and Larsson, EG May 2020 IEEE WIRELESS COMMUNICATIONS LETTERS 9(5), pp.581-585 Intelligent reflecting surfaces can improve the communication between a source and a destination. The surface contains metamaterial that is configured to "reflect" the incident wave from the source towards the destination. Two incompatible pathloss models have been used in prior work. In this letter, we derive the far-field pathloss model for the IRS-aided system. The results show that the proposed system can achieve a significant performance gain compared with the conventional IRS system. The results show that the proposed system can achieve a significant performance gain compared with the conventional IRS system. The results show that the proposed system can achieve a significant performance gain compared with the conventional IRS system. 全文链接 View full text Free Submitted Article From Repository ...	601 Citations 15 References Related records
10	Low-Complexity Hybrid Precoding in Massive Multiuser MIMO Systems Liang, L; Xu, W and Dong, XD Dec 2014 IEEE WIRELESS COMMUNICATIONS LETTERS 3(6), pp.653-656 Massive multiple-input multiple-output (MIMO) is envisioned to offer considerable capacity improvement, but at the cost of high complexity of the hardware. In this paper, we propose a low-complexity hybrid precoding scheme to approach the performance of the traditional baseband zero-forcing (ZF) precoding (referred to as full-complexity ZF precoding). The results show that the proposed system can achieve a significant performance gain compared with the conventional ZF precoding. The results show that the proposed system can achieve a significant performance gain compared with the conventional ZF precoding. The results show that the proposed system can achieve a significant performance gain compared with the conventional ZF precoding. 全文链接 Full Text at Publisher Free Submitted Article From Repository View Full Text on ProQuest ...	594 Citations 12 References Related records
11	Energy-Efficient Data Collection in UAV Enabled Wireless Sensor Network Zhan, C; Zeng, Y and Zhang, R Jun 2018 IEEE WIRELESS COMMUNICATIONS LETTERS 7(3), pp.328-331 In wireless sensor networks, utilizing the unmanned aerial vehicle (UAV) as a mobile data collector for the sensor nodes (SNs) is an energy-efficient technique to prolong the network lifetime. In this letter, considering a general fading channel model for the SN-UAV links, we jointly optimize the SNs' wakeup schedule and UAV's trajectory to maximize the network lifetime. The results show that the proposed system can achieve a significant performance gain compared with the conventional system. The results show that the proposed system can achieve a significant performance gain compared with the conventional system. The results show that the proposed system can achieve a significant performance gain compared with the conventional system. 全文链接 Full Text at Publisher ...	593 Citations 11 References Related records
12	Deep Learning-Based Channel Estimation for Beamspace mmWave Massive MIMO Systems He, HT; Wen, CK; (...); Li, GY Oct 2018 IEEE WIRELESS COMMUNICATIONS LETTERS 7(5), pp.852-855 Channel estimation is very challenging when the receiver is equipped with a limited number of radio-frequency (RF) chains in beamspace millimeter-wave massive multiple-input and multiple-output systems. To solve this problem, we exploit a learned denoising-based approximate message passing (LDAMP) network. This network can achieve a significant performance gain compared with the conventional system. The results show that the proposed system can achieve a significant performance gain compared with the conventional system. The results show that the proposed system can achieve a significant performance gain compared with the conventional system. 全文链接 Free Full Text From Publisher ...	588 Citations 10 References Related records
13	Reconfigurable Intelligent Surface Assisted UAV Communication: Joint Trajectory Design and Passive Beamforming Li, SX; Duo, B; (...); Di Renzo, M May 2020 IEEE WIRELESS COMMUNICATIONS LETTERS 9(5), pp.716-720 Thanks to the line-of-sight (LoS) transmission and flexibility, unmanned aerial vehicles (UAVs) effectively improve the throughput of wireless networks. Nevertheless, the LoS links are prone to severe deterioration by complex propagation environments, especially in urban areas. Reconfigurable intelligent surfaces (RISs), as a promising solution, can significantly improve the performance of LoS links. The results show that the proposed system can achieve a significant performance gain compared with the conventional system. The results show that the proposed system can achieve a significant performance gain compared with the conventional system. The results show that the proposed system can achieve a significant performance gain compared with the conventional system. 全文链接 View full text Free Submitted Article From Repository ...	563 Citations 13 References Related records
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Volume: 7 Issue: 1 Page: 73-84
DOI: 10.1109/TCCN.2020.3027695

Published MAR 2021

Indexed 2021-03-01

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Abstract An unmanned aerial vehicle (UAV)-aided mobile edge computing (MEC) framework is proposed, where several UAVs

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DOI: 10.1109/TVT.2019.2961993

PublishedFEB 2020

Indexed2020-02-01

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Abstract
This paper investigates an unmanned aerial vehicle (UAV)-enabled wireless communication system with energy harvesting, where the UAV transfers energy to the users in half duplex or full duplex, and the users harvest energy for data transmission to the UAV. We minimize the total energy consumption of the UAV while accomplishing the minimal data transmission requests of the users. The original optimization problem is decomposed into two

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Volume: 23 Issue: 9 Page: 1488-1492
DOI: 10.1109/LCOMM.2019.2924214

Published SEP 2019

Indexed 2019-09-01

Document Type Article

Abstract We investigate transmission optimization for intelligent reflecting surface (IRS) assisted multi-antenna systems from

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DOI: 10.1109/TCOMM.2017.2699194

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Abstract
The widely deployed cellular network, assisted with device-to-device (D2D) communications, can provide a promising solution to support efficient and reliable vehicular communications. Fast channel variations caused by high mobility in a vehicular environment need to be properly accounted for when designing resource allocation schemes for the D2D-enabled vehicular networks. In this paper, we perform spectrum sharing and power allocation

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Volume: 19 Issue: 8 Page: 5218-5233
DOI: 10.1109/TWC.2020.2990766

Published AUG 2020

Indexed 2020-08-01

Document Type Article

Abstract

Intelligent reflecting surfaces (IRSs) constitute a disruptive wireless communication technique capable of creating a controllable propagation environment. In this paper, we propose to invoke an IRS at the cell boundary of multiple cells to assist the downlink transmission to cell-edge users, whilst mitigating the inter-cell interference, which is a crucial issue in multicell communication systems. We aim for maximizing the weighted sum rate (WSR) of all users through jointly optimizing the active precoding matrices at the base stations (BSs) and the phase shifts at the IRS subject to each BS's power constraint and unit modulus constraint. Both the BSs and the users are equipped with multiple antennas, which enhances the spectral efficiency by exploiting the spatial multiplexing gain. Due to the non-convexity of the problem, we first reformulate it into an equivalent one, which is solved by using the block coordinate descent (BCD) algorithm, where the precoding matrices and phase shifts are alternately optimized. The optimal precoding matrices can be obtained in closed form, when fixing the phase shifts. A pair of efficient algorithms are proposed for solving the phase shift optimization problem, namely the Majorization-Minimization (MM) Algorithm and the Complex Circle Manifold (CCM) Method. Both algorithms are guaranteed to converge to at least locally optimal solutions. We also extend the proposed

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Published DEC 2014

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Abstract

Massive multiple-input multiple-output (MIMO) is envisioned to offer considerable capacity improvement, but at the cost of high complexity of the hardware. In this paper, we propose a low-complexity hybrid precoding scheme to approach the performance of the traditional baseband zero-forcing (ZF) precoding (referred to as full-complexity ZF), which is considered a virtually optimal linear precoding scheme in massive MIMO systems. The proposed hybrid precoding scheme, named phased-ZF (PZF), essentially applies phase-only control at the RF domain and then performs a low-dimensional baseband ZF precoding based on the effective channel seen from baseband. Heavily quantized RF phase control up to 2 bits of precision is also considered and shown to incur very limited degradation. The proposed scheme is simulated in both ideal Rayleigh fading channels and sparsely scattered millimeter wave (mmWave) channels, both achieving highly desirable performance.

Keywords

Author Keywords: [Massive MIMO](#); [hybrid precoding](#); [millimeter wave \(mmWave\) MIMO](#); [RF chain limitations](#)

Keywords Plus: [WIRELESS](#); [TRANSMISSION](#)

Author Information

Corresponding Address: Liang, Le (corresponding author)

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Chen, MZ; Yang, ZH; (...); Cui, SG

Jan 2021 | IEEE TRANSACTIONS ON WIRELESS COMMUNICATIONS 20(1), pp.269-283

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In this article, the problem of training federated learning (FL) algorithms over a realistic wireless network is studied. In the considered model, wireless users execute an FL algorithm while training their local FL models using their own data and transmitting the trained local FL models to a base station (BS) that generates a global FL mo ... Show more

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Tang, WK; Chen, MZ; (...); Cui, TJ

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Reconfigurable intelligent surfaces (RISs) comprised of tunable unit cells have recently drawn significant attention due to their superior capability in manipulating electromagnetic waves. In particular, RIS-assisted wireless communications have the great potential to achieve significant performance improvement and coverage enhancen ... Show more

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Yang, K; Jiang, T; (...); Ding, Z

Mar 2020 | IEEE TRANSACTIONS ON WIRELESS COMMUNICATIONS 19(3), pp.2022-2035

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Yang, ZH; Chen, MZ; (...); Shikh-Bahaei, M

Mar 2021 | IEEE TRANSACTIONS ON WIRELESS COMMUNICATIONS 20(3), pp.1935-1949

In this paper, the problem of energy efficient transmission and computation resource allocation for federated learning (FL) over wireless communication networks is investigated. In the considered model, each user exploits limited local computational resources to train a local FL model with its collected data and, then, sends the ... Show more

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Al-Hourani, A; Kandeepan, S and Lardner, S

Dec 2014 | IEEE WIRELESS COMMUNICATIONS LETTERS 3(6), pp.569-572

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Ye, H; Li, GY and Juang, BH

Feb 2018 | IEEE WIRELESS COMMUNICATIONS LETTERS 7(1), pp.114-117

This letter presents our initial results in deep learning for channel estimation and signal detection in orthogonal frequency-division multiplexing (OFDM) systems. In this letter, we exploit deep learning to handle wireless OFDM channels in an end-to-end manner. Different from existing OFDM receivers that first estimate channel state infor ... Show more

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Wen, CK; Shih, WT and Jin, S

Oct 2018 | IEEE WIRELESS COMMUNICATIONS LETTERS 7(5), pp.748-751

In frequency division duplex mode, the downlink channel state information (CSI) should be sent to the base station through feedback links so that the potential gains of a massive multiple-input multiple-output can be exhibited. However, such a transmission is hindered by excessive feedback overhead. In this letter, we use deep l ... Show more

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4 Intelligent Reflecting Surface Versus Decode-and-Forward: How Large Surfaces are Needed to Beat Relaying?



Björnson, E; Özdogan, Ö and Larsson, EG

Feb 2020 | IEEE WIRELESS COMMUNICATIONS LETTERS 9(2), pp.244-248

The rate and energy efficiency of wireless channels can be improved by deploying software-controlled metasurfaces to reflect signals from the source to the destination, especially when the direct path is weak. While previous works mainly optimized the reflections, this letter compares the new technology with classic decode-and-f ... Show more

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5 Secure Wireless Communication via Intelligent Reflecting Surface



Cui, M; Zhang, GC and Zhang, R

Oct 2019 | IEEE WIRELESS COMMUNICATIONS LETTERS 8(5), pp.1410-1414

An intelligent reflecting surface (IRS) can adaptively adjust the phase shifts of its reflecting units to strengthen the desired signal and/or suppress the undesired signal. In this letter, we investigate an IRS-aided secure wireless communication system where a multi-antenna access point (AP) sends confidential messages to a single-ant ... Show more

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6 3-D Placement of an Unmanned Aerial Vehicle Base Station (UAV-BS) for Energy-Efficient Maximal Coverage



Alzenad, M; El-Keyi, A; (...); Yanikomeroglu, H

Aug 2017 | IEEE WIRELESS COMMUNICATIONS LETTERS 6(4), pp.434-437

Unmanned aerial vehicle mounted base stations (UAV-BSs) can provide wireless services in a variety of scenarios. In this letter, we propose an optimal placement algorithm for UAV-BSs that maximizes the number of covered users using the minimum transmit power. We decouple the UAV-BS deployment problem in the vertical and ... Show more

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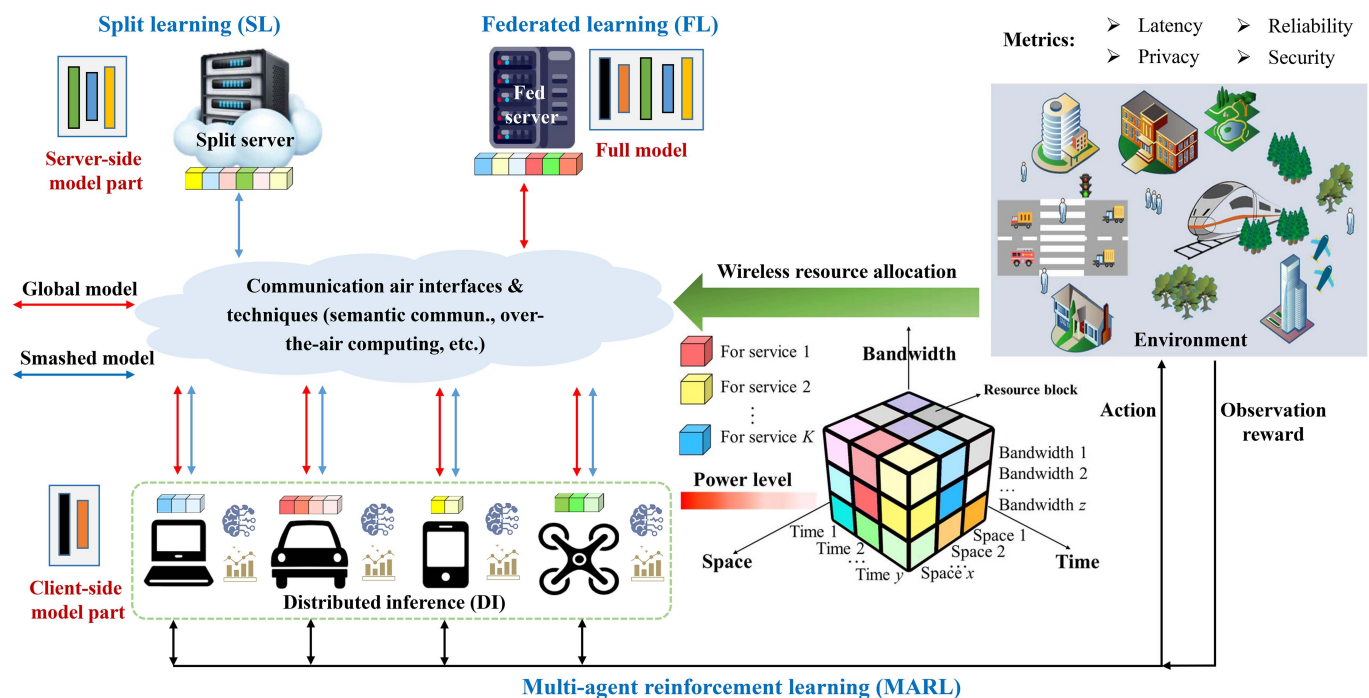
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